

Digital Electronics

Number System

Decimal Base = 10	Binary Base = 2	Octal Base = 8	Hexa-decimal Base = 16
0	0	0	0 to 9
1	1	1	10 - A
2		2	11 - B
3		3	12 - C
4		4	13 - D
5		5	14 - E
6		6	15 - F
7		7	
8			
9			

Binary Addition

$$\begin{array}{r} \overset{\textcircled{0}}{1} \overset{\textcircled{0}}{0} 1 1 \\ + 1 0 0 1 \\ \hline 1 0 1 0 0 \end{array}$$

$$\begin{array}{r} \overset{\textcircled{1}}{1} \overset{\textcircled{1}}{1} 0 1 \\ + 1 1 0 1 \\ \hline 1 1 0 1 0 \end{array}$$

1's Complement

$$\neg(1 0 0 1) = 0 1 1 0$$

$$1's \text{ complement of } 1101 = 0010$$

2's Complement

$$\boxed{2's \text{ complement} = 1's \text{ complement} + 1}$$

Q. 1010

$$\begin{array}{r} 1's \text{ complement} = 0101, \quad \overset{\textcircled{1}}{0} 1 0 1 \\ + 0 0 0 1 \\ \hline 0 1 1 0 \end{array}$$

$$2's \text{ complement of } 1010 = 0110.$$

Q. 10101101

$$\begin{array}{r} 1's \text{ complement} = 01010010, \quad 0 1 0 1 0 0 1 0 \\ + 0 0 0 0 0 0 0 1 \\ \hline 0 1 0 1 0 0 1 1 \end{array}$$

$$2's \text{ complement of } 10101101 = 01010011$$

Binary Subtraction Using 1's Complement

$$(10)_{10} - (8)_{10}$$

$$\begin{array}{r} 1010 \longrightarrow 1010 \\ 1000 \xrightarrow{1's\ comp} 0111 \end{array} \longrightarrow \begin{array}{r} \overset{0}{1} \overset{0}{0} 10 \\ + 0111 \\ \hline \boxed{1} 0001 \\ 0001 \\ + 0001 \\ \hline 0010 \end{array}$$

$$(10)_{10} - (8)_{10} = (0010)_2 = (2)_{10}.$$

Binary Subtraction Using 2's Complement

$$(10)_{10} - (8)_{10}$$

$$\begin{array}{r} 1010 \longrightarrow 1010 \\ 1000 \xrightarrow{2's\ comp} 1000 \end{array} \longrightarrow \begin{array}{r} 1010 \\ + 1000 \\ \hline \cancel{X} \boxed{1} 0010 \end{array}$$

$$(10)_{10} - (8)_{10} = (2)_{10}.$$

$$(10)_{10} - (2)_{10}$$

$$\begin{array}{r} 1010 \longrightarrow 1010 \\ 0010 \xrightarrow{2's\ comp} 1110 \end{array} \longrightarrow \begin{array}{r} \overset{0}{1} \overset{0}{0} 10 \\ + 1110 \\ \hline \cancel{X} \boxed{1} 1000 \end{array}$$

$$(10)_{10} - (2)_{10} = (8)_{10}.$$

DeMorgan's Theorem

There are two theorems:-

1. Complement of a product is equal to sum of complements.

$$\overline{A \cdot B} = \bar{A} + \bar{B}$$

$$\overline{A \cdot B \cdot C} = \bar{A} + \bar{B} + \bar{C}$$

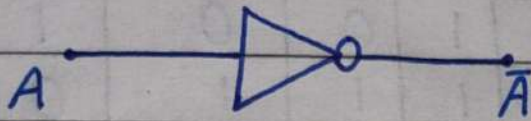
2. Complement of a sum is equal to product of complements.

$$\overline{A + B} = \bar{A} \cdot \bar{B}$$

$$\overline{A + B + C} = \bar{A} \cdot \bar{B} \cdot \bar{C}$$

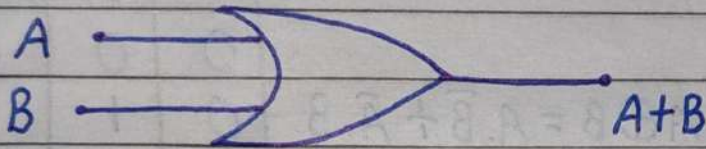
Basic/Standard Gates

NOT



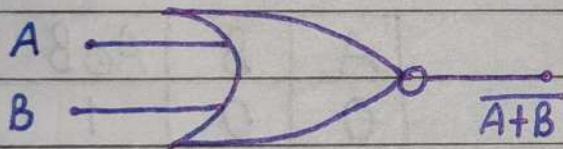
A	$\bar{A}$
0	1
1	0

OR



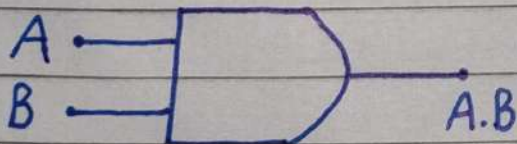
A	B	$A+B$
0	0	0
0	1	1
1	0	1
1	1	1

NOR



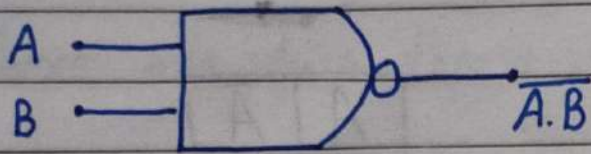
A	B	$A+B$	$\overline{A+B}$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

AND



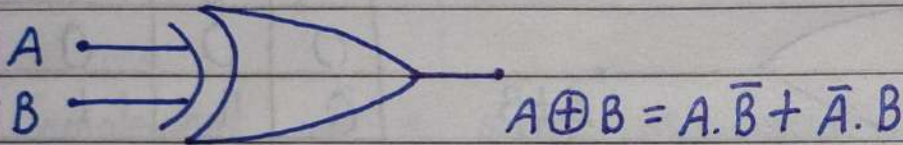
A	B	$A.B$
0	0	0
0	1	0
1	0	0
1	1	1

NAND



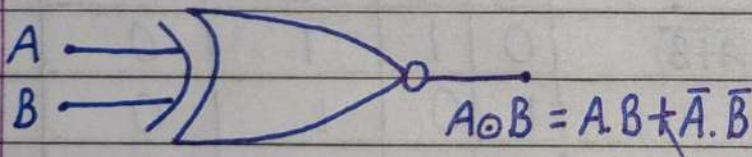
A	B	$A.B$	$\overline{A.B}$
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

Imp: XOR (Exclusive OR)



A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

Imp: X-NOR (Exclusive NOR)



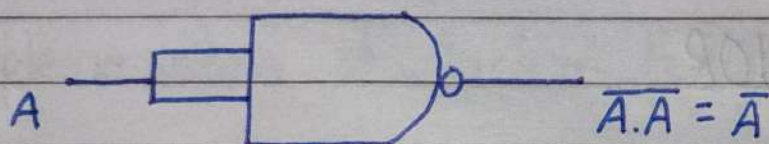
A	B	$A \odot B$
0	0	1
0	1	0
1	0	0
1	1	1

Imp # Universal Gates

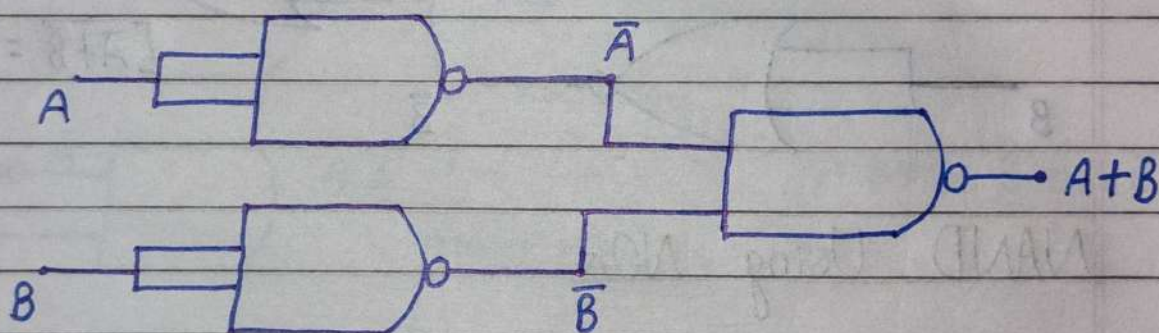
NAND & NOR gates are called as universal gates because any boolean function can be implemented/realized using these (NAND/NOR) gates.

Note: $A.A.A = A$ & $A+A+A = A$.

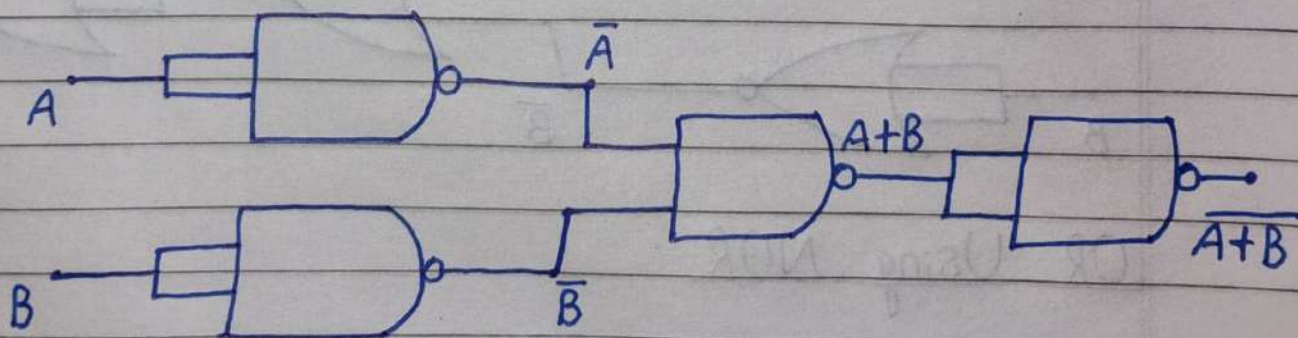
NOT Using NAND



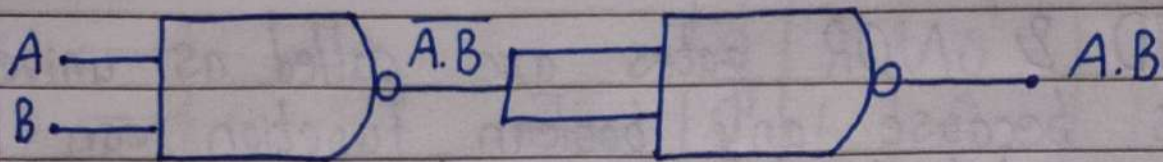
OR Using NAND



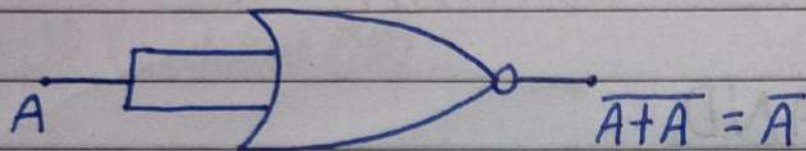
NOR Using NAND



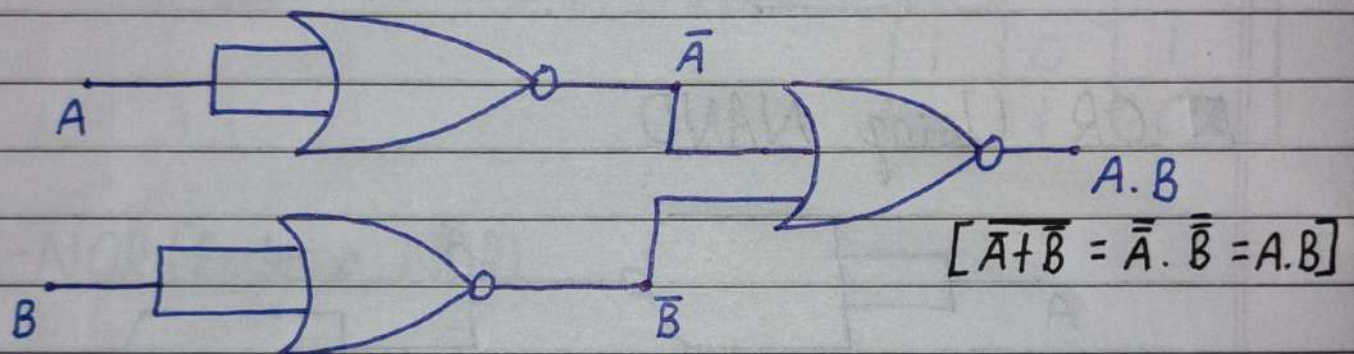
AND Using NAND



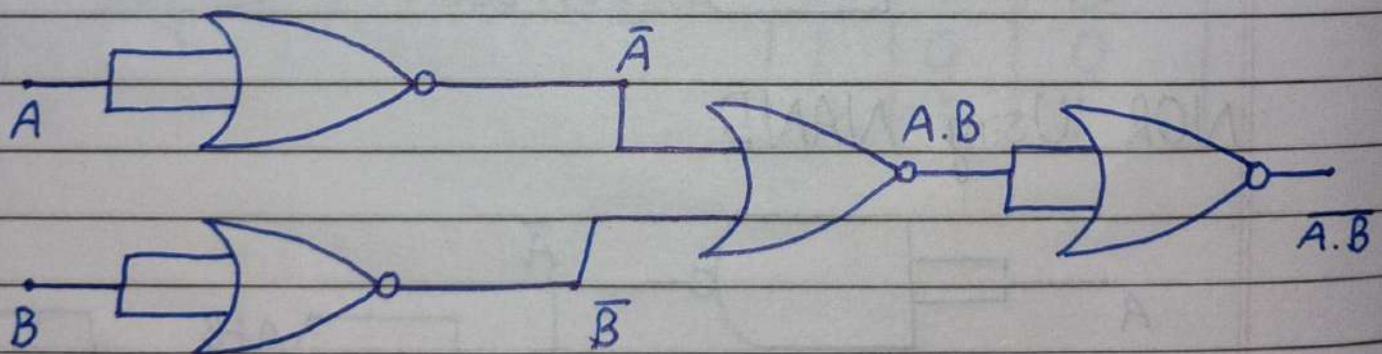
NOT Using NOR



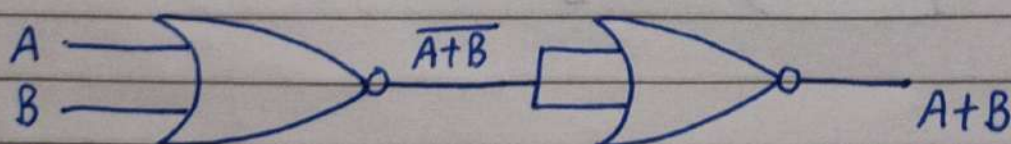
AND Using NOR



NAND Using NOR



OR Using NOR



Types of Boolean Functions

1. Sum of Product (SOP)

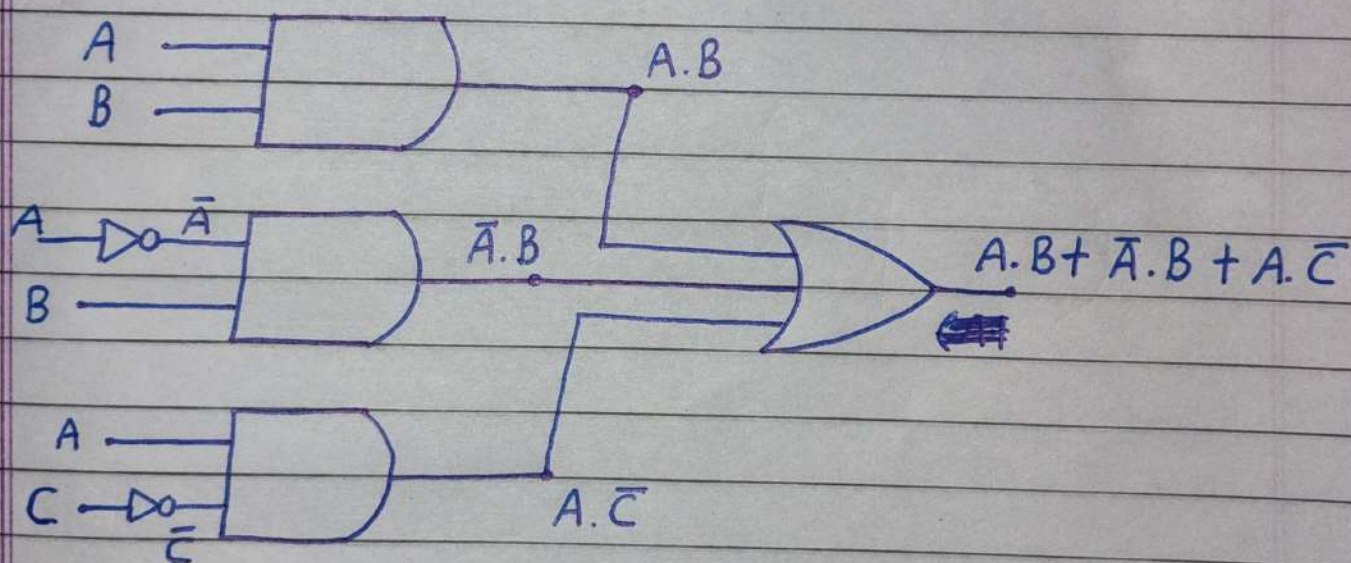
$$F = A.B + \bar{A}.B + A.\bar{C}$$

2. Product of Sum (POS)

$$F = (A+B).(\bar{A}+C).(A+\bar{C})$$

Implementation of Boolean Function Using Basic Gates

$$F = A.B + \bar{A}.B + A.\bar{C}$$



Types of Boolean Functions

1. Sum of Product (SoP)

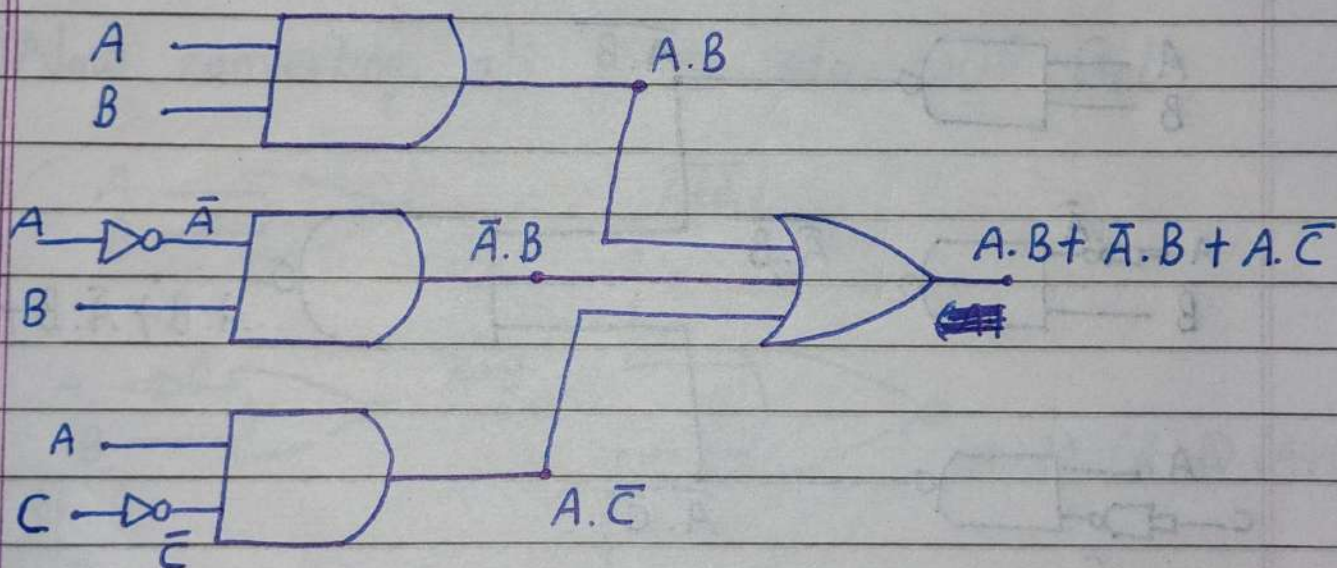
$$F = A.B + \bar{A}.B + A.\bar{C}$$

2. Product of Sum (PoS)

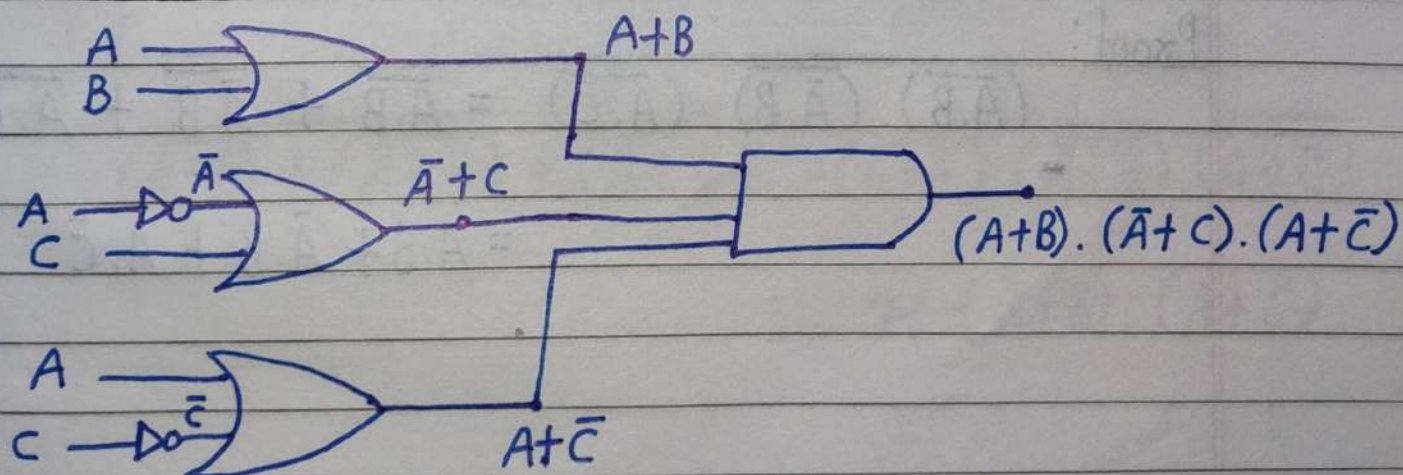
$$F = (A+B).(\bar{A}+C).(A+\bar{C})$$

Implementation of Boolean Function Using Basic Gates

$$F = A.B + \bar{A}.B + A.\bar{C}$$

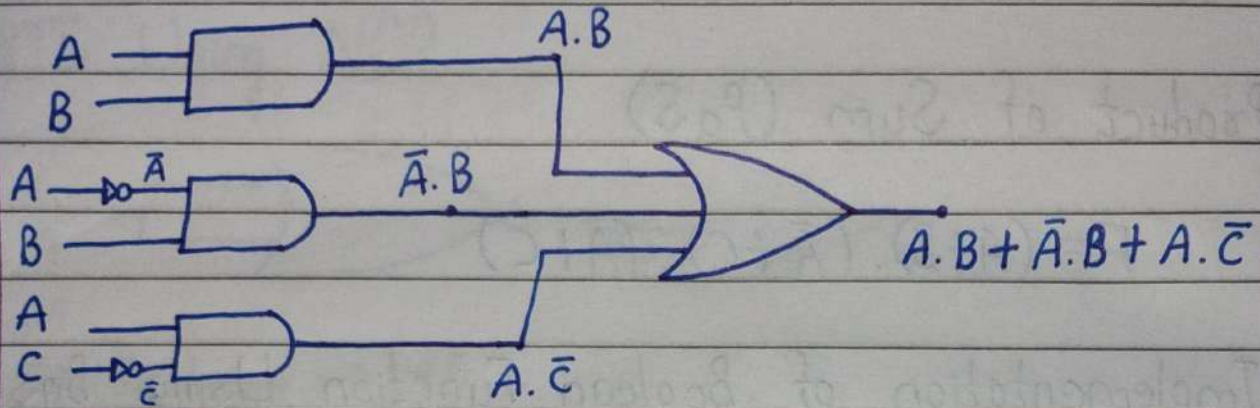


$$F = (A+B).(\bar{A}+C).(A+\bar{C})$$

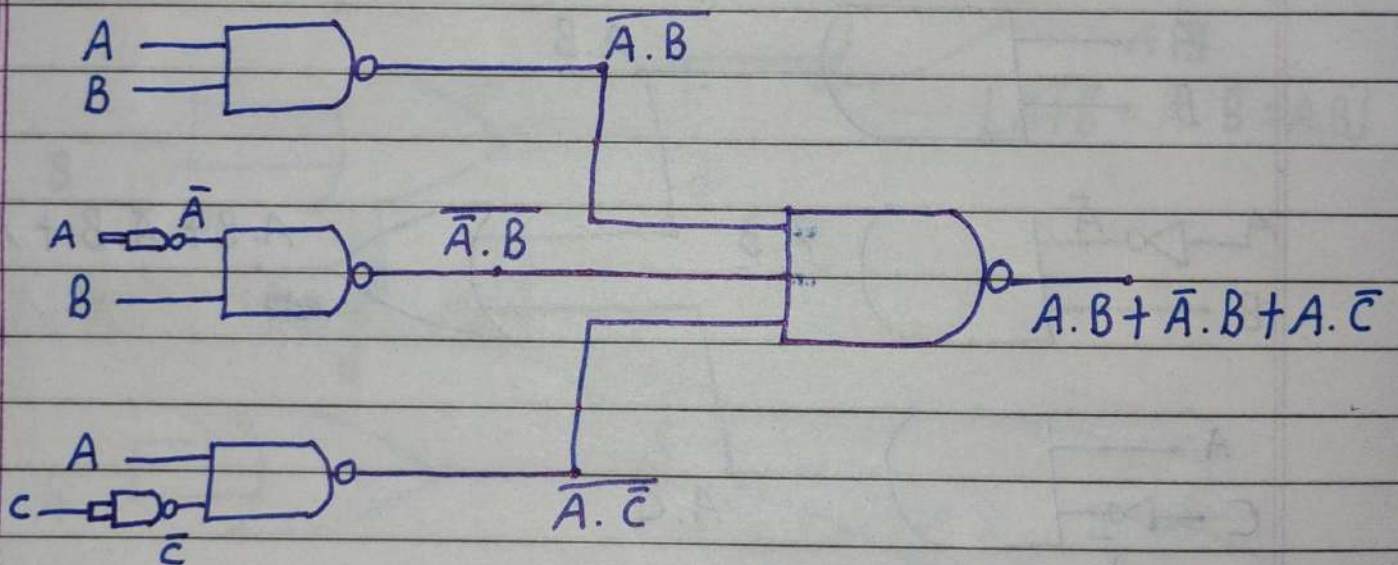


Implementation of Boolean Function Using NAND Gate Only (Universal Gate)

$$F = A.B + \bar{A}.B + A.\bar{C}$$



Now converting all gates into NAND Gate,



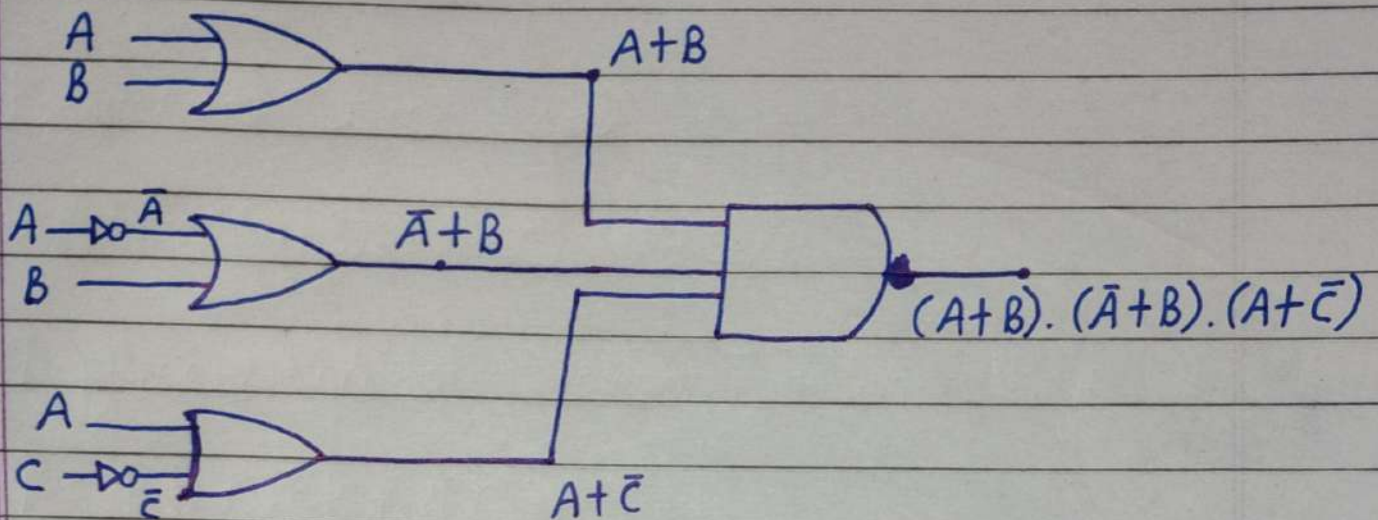
Proof:

$$\overline{(\overline{A.B})} . \overline{(\overline{\bar{A}.B})} . \overline{(\overline{A.\bar{C}})} = \overline{\overline{A.B}} + \overline{\overline{\bar{A}.B}} + \overline{\overline{A.\bar{C}}}$$

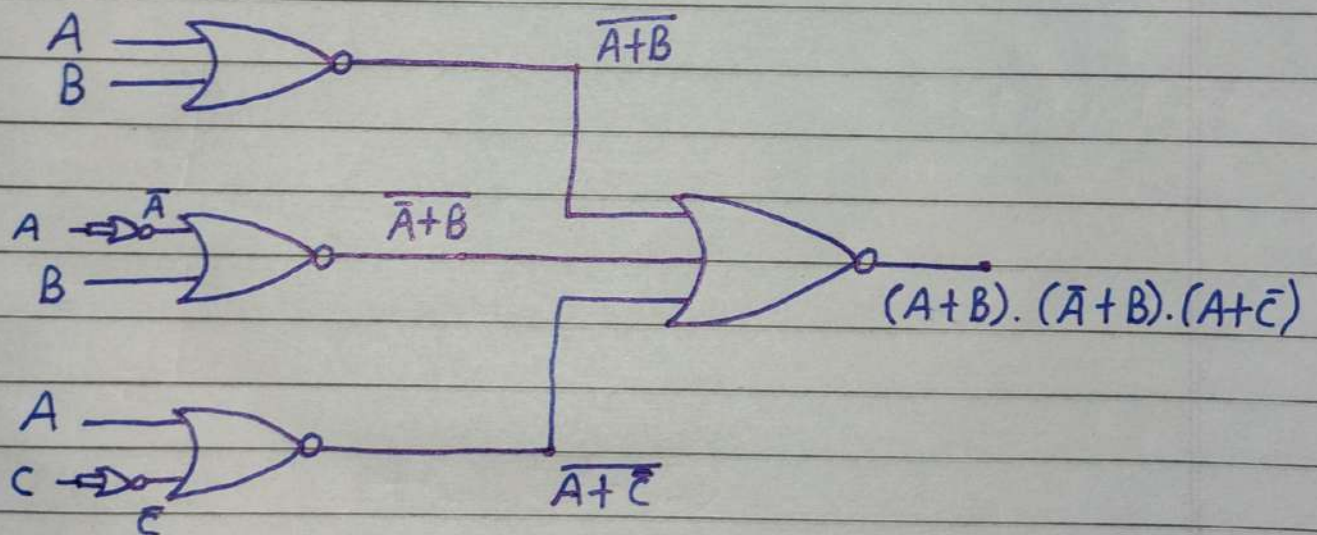
$$= A.B + \bar{A}.B + A.\bar{C}$$

Implementation of Boolean Function Using NOR Gate Only (Universal Gate)

$$F = (A+B).(\bar{A}+B).(A+\bar{C})$$



Now converting all gates into NOR Gate,



Proof:

$$\begin{aligned} \overline{(\bar{A}+B)} + \overline{(\bar{A}+B)} + \overline{(A+\bar{C})} &= \overline{(\bar{A}+B)}. \overline{(\bar{A}+B)}. \overline{(A+\bar{C})} \\ &= (A+B).(\bar{A}+B).(A+\bar{C}) \end{aligned}$$